# The Irreducible Boundary of a Bounded Resolvable Space: Individuation, Boundary Thickness, and the Resolution Floor

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**Abstract**

We study *individuation* in a bounded resolvable space: the act of separating a part from the whole when no element of the space is distinguished in advance. We prove that, in a bounded space carrying a finite resolution, individuation can proceed only by negation—a part is fixed as the complement of everything it is not—and that every such separation deposits content of strictly positive measure into a boundary. This content, the *resolution floor* $\beta > 0$, is the central invariant of the paper. We establish a *Boundary–Thickness Theorem* (no separator between a part and its complement has zero content), a *Non-Instantaneity Theorem* (a separation that yields a genuine part cannot be completed in zero steps, and what the process leaves untouched is an irreducible residue), and a *Detectability Theorem* (a part is distinguishable exactly when its own scale exceeds its boundary thickness; sub-floor parts are engulfed by their separator). The classical sorites is a corollary: there is no sharp count at which an aggregate becomes distinguishable, because the separating boundary has positive width and distinguishability is emergence from that width. We give three independent derivations of the floor—representational, cardinality-theoretic, and a cost argument by which compressing a separator to zero content diverges—and prove they bound the same constant. We then separate two relations a region may bear: being *located* by an agent that individuates it, which is costly and whose cost grows with the logarithm of the region's rarity, and being *visited* by a measure-preserving traversal, which is free and scales with rarity only in frequency; from this asymmetry we classify three invalid inferences that transfer a cost of locating onto being reached. We then identify the deepest obstruction: to *verify* that a part is what it is requires deciding its membership in the very collection that constitutes it (its negation), a self-referential question with no consistent value. The irreducible residue is therefore not a locatable set but a self-referential impossibility, proved here by a self-contained diagonal argument; it cannot be exhibited, only bounded below by $\beta$. From this we obtain a *Non-Return Theorem*—the count of committed separations is monotone non-decreasing under every traversal, so no process returns to a prior individuation state, and resemblance never entails identity—and an *Inquiry–Cessation Theorem*: thinning a separator toward zero content has diverging cost and bounded return, so a finite agent halts at the floor, and knowledge thereafter attaches to bounded wholes and never to their sub-floor parts. Every claim is proved from stated definitions; uncertain material has been omitted.

**Keywords:** bounded resolvable space; individuation; negation; topological boundary; boundary thickness; resolution floor; sorites paradox; vagueness; self-reference; diagonal argument; irreducible residue; monotone individuation.

## Contents

# 1 Introduction

## 1.1 The problem of individuation without a selector

A bounded space contains no part until one is made. To speak of a part $A$ is to have separated $A$ from its complement; but if no element of the space is distinguished in advance—if there is no privileged point, no selector, no prior "main actor"—then the first question is how a part comes to be individuated at all. This paper takes that question as primary and answers it with a single device, negation: a part is fixed not by being chosen but by being marked off from *everything it is not.* The complement requires no selector, because it is the whole minus the part and is given as soon as the part is; and the part is recovered as the one thing the complement omits. We prove that this is essentially the *only* selector-free individuation (Section 4), and we trace its consequences.

## 1.2 Thesis

Under one hypothesis—that the space is bounded and finitely resolvable (Axiom 1)—we prove the following chain.

(i) *Individuation is by negation.* A part is the complement of its negation; this is the unique selector-free way to obtain a determinate part from an undifferentiated whole.

(ii) *Every separation has positive thickness.* No part can be cut from its complement by a separator of zero content. The separator carries content $\geq \beta > 0$, the resolution floor (Section 3).

(iii) *Separation is non-instantaneous, and leaves a residue.* A separation completed in zero steps yields no part; an actual separation occupies the process and leaves an untouched remainder (Section 4).

(iv) *Detection is emergence from the boundary.* A part is distinguishable exactly when its scale exceeds its own boundary thickness; below that scale it is engulfed (Section 6). The sorites paradox is a corollary.

(v) *Locating is priced; being reached is not.* The cost of individuation is paid by singling a region out, and scales with the logarithm of how rare the region is; it is not paid by a measure-preserving traversal that merely passes through the region, which does so freely and only less often when the region is rarer (Section 7).

(vi) *Verification is self-referentially obstructed.* To verify a part is to decide its membership in the collection that constitutes it; this has no consistent value, so the residue is irreducible—bounded below by $\beta$ yet not exhibitable as a set (Section 8).

(vii) *Individuation does not return.* The count of committed separations is monotone, so no process recovers a prior individuation state; knowledge attaches to bounded wholes and halts at the floor (Section 9).

## 1.3 On the word "knowledge"

We use *knowledge* in a single, fully operational sense throughout: to know a part is to be able to verify it, and to verify a part is to decide, against its complement, that it is the part it is. No other notion of knowledge is invoked, and nothing in the paper depends on any informal reading of the word. With this definition, "the boundary of knowledge" is literally the boundary of the verifying separation, and the theorems about boundaries are theorems about knowledge in this exact sense.

## 1.4 Why "thickness"

The everyday and the formal meet at the sorites. An aggregate built one unit at a step is, at first, not distinguishable as an aggregate; at some size it is; yet no single unit is the one that makes it so. Classically this is a paradox. We resolve it by inverting the reading: a sub-threshold collection is indistinguishable not because it is small in the absolute but because its *separating boundary is thick relative to it*—the layered separator one must resolve to mark it off from everything else is wider than the collection, so it is engulfed by its own boundary and never stands out. It becomes distinguishable when it is large compared with that separator. There is no critical unit because distinguishability is emergence across a band of positive width, not the crossing of a point; the width of the band is the floor $\beta$ (Theorem 6.3, Theorem 6.6).

## 1.5 Why "bounded"

Boundedness is the source of the floor, not a convenience. On an unbounded space a part may be separated from its complement by an arbitrarily thin frontier, and structure may be resolved to arbitrary depth at fixed cost. It is the finiteness of the space, together with the finiteness of the resolution any embedded agent can realise, that forbids the zero-thickness cut. We make the hypothesis exactly strong enough to force the floor and no stronger, and we mark, at each step, where boundedness is used (Theorem 3.10).

## 1.6 Relation to existing work

The paper is pure mathematics: topology and measure theory (Engelking, 1995; Federer, 1969; Halmos, 1950; Hurewicz and Wallman, 1941; Kelley, 1955; Mattila, 1995; Munkres, 2000), order and fixed-point theory (Birkhoff, 1967; Davey and Priestley, 2002; Tarski, 1955), and self-reference (Cantor, 1891; Gödel, 1931; Lawvere, 1969; Russell, 1903; Tarski, 1936). The sorites and vagueness have substantial literatures—degree and fuzzy theories (Zadeh, 1965), supervaluation (Fine, 1975; Keefe, 2000), epistemicism (Williamson, 1994), and the classical analysis of Black (1937); see Hyde and Raffman (2018) for a survey. Our treatment is none of these: it is not a logic of partial truth, nor a claim that a sharp boundary exists but is unknown, but a *geometric* statement that the separator is a set of positive content whose verification is self-referentially obstructed. The self-reference results we use are proved here in self-contained form; the classical results of Cantor (1891); Russell (1903); Tarski (1936) are cited only as the shape the argument shares. A reasonable reader will recognise that the abstract hypotheses below are satisfied by a range of concrete systems—and that several of the problems we phrase abstractly, about rare regions costly to single out yet freely reached by a traversal (Section 7), have well-known concrete instances; we make no such identification and prove everything of the abstract space alone, using only quantities (measure, log-measure, probability, visiting frequency, cost) that are defined within measure theory and information theory (Cover and Thomas, 2006; Shannon, 1948).

### 1.7 Method

Every substantive statement is a definition, a theorem proved here, or a citation of a standard mathematical result compared against such a theorem. Where a tempting claim could not be proved at this standard, it has been removed rather than weakened. The development is uniform: a single hypothesis, a single invariant $\beta$, and a chain of theorems that return to it from six directions—topological, representational, cardinality-theoretic, economic, self-referential, and order-theoretic—each shown to bound or price the same constant.

### 1.8 Organisation

Section 2 fixes the bounded resolvable space and its hypothesis. Section 4 proves that individuation must be by negation and that it is non-instantaneous, yielding the residue. Section 3 proves the Boundary–Thickness Theorem. Section 5 gives the three agreeing derivations of the floor. Section 6 proves the Detectability Theorem and resolves the sorites. Section 7 distinguishes locating a region from a traversal's visiting it, proves the asymmetry of their costs, and classifies three invalid transfers between them. Section 8 proves the self-referential obstruction to verification and characterises the irreducible residue. Section 9 proves the Non-Return and Inquiry–Cessation Theorems. Section 10 states scope and concludes.

## 2 The setting: a bounded resolvable space

We work throughout in a metric–measure space. The objects of the theory are regions (subsets), their separators (boundaries), and partitions; the single hypothesis constrains the ambient space to be bounded and finitely resolvable.

### 2.1 Ambient space

**Definition 2.1** (Resolvable space)**.** A *resolvable space* is a triple $(\Omega, d, \mu)$ where $(\Omega, d)$ is a metric space and $\mu$ is a Borel measure on $\Omega$ that is positive on non-empty open sets and finite on bounded sets. We write $\mathcal{L}$ for Lebesgue measure when $\Omega \subseteq \mathbb{R}^m$, and $\mathcal{H}^s$ for $s$-dimensional Hausdorff measure (Federer, 1969; Mattila, 1995).

**Definition 2.2** (Region, complement, separator)**.** A *region* is a Borel set $A \subseteq \Omega$ with $\mu(A) > 0$ and $\mu(\Omega \setminus A) > 0$ (so both $A$ and its complement are non-negligible). Its *topological boundary* is $\partial A = \mathrm{cl}(A) \cap \mathrm{cl}(\Omega \setminus A)$. A *separator* of $A$ is any Borel set $\Sigma$ with $\Omega \setminus \Sigma = U_A \sqcup U_{A^c}$ a disjoint union of two non-empty relatively open sets with $A \setminus \Sigma \subseteq U_A$ and $(\Omega \setminus A) \setminus \Sigma \subseteq U_{A^c}$. A separator is *sharp* if $\mu(\Sigma) = 0$.

*Remark* 2.3. $\partial A$ is the smallest closed separator in the topological sense; a general separator may be thicker. The question the paper turns on is whether a *sharp* separator (one of zero measure) can exist for the regions that an embedded agent actually uses. The answer, under Axiom 1, is no.

### 2.2 Partitions and resolution

**Definition 2.4** (Finite measurable partition)**.** A *finite measurable partition* of $\Omega$ is a finite family $\mathscr{P} = \{\mathsf{c}_1, \ldots, \mathsf{c}_N\}$ of pairwise-disjoint Borel sets with $\bigcup_i \mathsf{c}_i = \Omega$ and $\mu(\mathsf{c}_i) > 0$ for each $i$. A partition $\mathscr{P}'$ *refines* $\mathscr{P}$, written $\mathscr{P}' \preceq \mathscr{P}$, if every cell of $\mathscr{P}'$ is contained in a cell of $\mathscr{P}$.

**Definition 2.5** (Resolution)**.** A resolvable space *carries resolution* $\delta > 0$ for a partition $\mathscr{P}$ if every cell $\mathsf{c} \in \mathscr{P}$ satisfies $\mathrm{diam}(\mathsf{c}) \geq \delta$; equivalently, no two points closer than $\delta$ are forced into distinct cells. The resolution is the finest scale at which the partition distinguishes points.

*Remark* 2.6 (Resolution is an agent-relative datum). Resolution is not a property of $\Omega$ alone but of the partition an agent can realise. Two agents resolving the same space may carry different $\delta$. The floor we derive will inherit this relativity: its magnitude depends on $\delta$, but its strict positivity will not.

### 2.3 The hypothesis

**Axiom 1** (Bounded resolvable space)**.** The space $(\Omega, d, \mu)$ satisfies:

(BRS1) **Boundedness.** $\Omega$ is bounded and $0 < \mu(\Omega) < \infty$; moreover $\mathrm{cl}(\Omega)$ is compact.

(BRS2) **Finite resolution.** Every partition realisable by an embedded agent carries some resolution $\delta > 0$; there is no partition into cells of arbitrarily small diameter at fixed, finite descriptive cost.

(BRS3) **Hierarchical refinability.** There is a sequence of finite measurable partitions $\mathscr{P}_0 \succeq \mathscr{P}_1 \succeq \mathscr{P}_2 \succeq \cdots$, each refining the last, generating the Borel $\sigma$-algebra modulo $\mu$-null sets; each $\mathscr{P}_n$ carries a resolution $\delta_n > 0$.

*Remark* 2.7 (Status of the hypothesis). (BRS1) is the substantive geometric content: the accessible space is finite in extent and measure, and its closure is compact (so the extreme-value and finite-subcover arguments below apply). (BRS2) is the finitude of any embedded agent: an agent with finite descriptive resources cannot specify cells of vanishing diameter—this is what forbids the limiting zero-thickness cut. (BRS3) states that finer partitions exist and exhaust the measurable structure, but each *individual* partition is still finite-resolution; refinement is unbounded in the limit but bounded at every finite stage. The three clauses are exactly what is needed for the Boundary–Thickness Theorem and nothing more.

**Lemma 2.8** (Finiteness of cell count)**.** *Under* (BRS1) *and* (BRS2)*, every realisable partition* $\mathscr{P}$ *has finitely many cells, and the count is bounded by* $N(\mathscr{P}) \le \mu(\Omega)/\mu_{\min}(\mathscr{P})$*, where* $\mu_{\min}(\mathscr{P}) = \min_i \mu(\mathsf{c}_i) > 0$*.*

*Proof.* By (BRS2) each cell has $\mathrm{diam}(\mathsf{c}_i) \ge \delta$, and by (BRS1) $\mu(\mathsf{c}_i) > 0$ with the cells disjoint and of finite total measure $\mu(\Omega) < \infty$. If the count were infinite, then $\sum_i \mu(\mathsf{c}_i) = \mu(\Omega) < \infty$ would force $\inf_i \mu(\mathsf{c}_i) = 0$, contradicting that finitely-described cells carry a uniform positive lower measure bound under finite resolution. Hence the count is finite, and $N(\mathscr{P})\,\mu_{\min}(\mathscr{P}) \le \sum_i \mu(\mathsf{c}_i) = \mu(\Omega)$ gives the bound. □

## 3 The Boundary–Thickness Theorem

We now prove the central structural fact: a bounded, finitely-resolved agent cannot separate a region from its complement by a set of zero content. Every separator it can realise has positive content, and that content is bounded below by a constant determined by the resolution—the floor.

### 3.1 Operational separators

The topological boundary $\partial A$ can have measure zero (e.g. a half-space in $\mathbb{R}^m$). The theorem is not about $\partial A$ in the ambient continuum; it is about the separators a finite-resolution agent can actually realise, which must be *unions of cells*.

**Definition 3.1** (Realisable region and separator)**.** Fix a partition $\mathscr{P}$ with resolution $\delta$. A region $A$ is $\mathscr{P}$*-realisable* if it is a union of cells of $\mathscr{P}$. A $\mathscr{P}$*-separator* of a $\mathscr{P}$-realisable $A$ is the set of cells of $\mathscr{P}$ that meet both $A$ and $\Omega \setminus A$ in the closure—formally,

$$\Sigma_{\mathscr{P}}(A) \;=\; \bigcup\{\mathsf{c} \in \mathscr{P} :\; \mathrm{cl}(\mathsf{c}) \cap \mathrm{cl}(A) \neq \varnothing \text{ and } \mathrm{cl}(\mathsf{c}) \cap \mathrm{cl}(\Omega \setminus A) \neq \varnothing\}.$$

Because a finite-resolution agent can only point to cells, $\Sigma_{\mathscr{P}}(A)$ is the thinnest separator it can name.

*Remark* 3.2. The shift from $\partial A$ to $\Sigma_{\mathscr{P}}(A)$ is the whole content of "operational." A continuum agent with unlimited resolution might use $\partial A$ of measure zero; a agent bound by (BRS2) cannot, because it cannot name sub-cell sets. The theorem quantifies the price of that limitation.

### 3.2 Non-triviality of the separator

**Lemma 3.3** (The realisable separator is non-empty)**.** *Let $A$ be $\mathscr{P}$-realisable with $\mu(A) > 0$ and $\mu(\Omega \setminus A) > 0$, and suppose $\Omega$ is connected. Then $\Sigma_{\mathscr{P}}(A) \neq \varnothing$.*

*Proof.* Suppose $\Sigma_{\mathscr{P}}(A) = \varnothing$. Then no cell meets the closures of both $A$ and $\Omega \setminus A$; hence $\mathrm{cl}(A)$ and $\mathrm{cl}(\Omega \setminus A)$ are disjoint unions of whole cells whose closures do not touch. Writing $U = \mathrm{int}(\mathrm{cl}A)$ and $V = \mathrm{int}(\mathrm{cl}(\Omega \setminus A))$, the sets $U, V$ are non-empty (each contains a cell of positive measure), open, disjoint, and cover $\Omega$ up to a $\mu$-null set; if their closures do not meet, $U \sqcup V$ is a separation of $\Omega$ into two non-empty clopen pieces, contradicting connectedness. Hence $\Sigma_{\mathscr{P}}(A) \neq \varnothing$. $\square$

*Remark* 3.4 (Connectedness). Theorem 3.3 uses connectedness of $\Omega$. If $\Omega$ is disconnected, a region can coincide with a component and have empty separator; but then $A$ was not genuinely *divided* from its complement—it was given. The theorem concerns acts of division *within* a connected whole, which is the case of interest (an object carved out of one world). We assume connectedness hereafter and note it where used.

### 3.3 The thickness lower bound

**Definition 3.5** (Boundary thickness)**.** The *boundary thickness* of a $\mathscr{P}$-realisable region $A$ is the measure of its realisable separator,

$$\beta_{\mathscr{P}}(A) \;:=\; \mu(\Sigma_{\mathscr{P}}(A)).$$

**Theorem 3.6** (Boundary–Thickness Theorem)**.** *Let $(\Omega, d, \mu)$ be a connected bounded resolvable space satisfying Axiom 1, and let $\mathscr{P}$ be a realisable partition of resolution $\delta$ with minimum cell measure $\mu_{\min} > 0$. Then for every $\mathscr{P}$-realisable region $A$ with $\mu(A) > 0$ and $\mu(\Omega \setminus A) > 0$,*

$$\boxed{\beta_{\mathscr{P}}(A) \;\geq\; \mu_{\min} \;>\; 0.}$$

*In particular no realisable separator is sharp: $\mu(\Sigma_{\mathscr{P}}(A)) = 0$ is impossible.*

*Proof.* By Theorem 3.3, $\Sigma_{\mathscr{P}}(A)$ contains at least one cell $\mathsf{c}^* \in \mathscr{P}$. By (BRS2) and Theorem 2.8 every cell has $\mu(\mathsf{c}) \geq \mu_{\min} > 0$. Since $\Sigma_{\mathscr{P}}(A)$ is a union of cells and contains $\mathsf{c}^*$,

$$\beta_{\mathscr{P}}(A) = \mu(\Sigma_{\mathscr{P}}(A)) \;\geq\; \mu(\mathsf{c}^*) \;\geq\; \mu_{\min} \;> 0.$$

A sharp separator would have $\mu(\Sigma_{\mathscr{P}}(A)) = 0 < \mu_{\min}$, a contradiction. $\square$

**Definition 3.7** (Resolution floor)**.** The *resolution floor* of a agent realising partition $\mathscr{P}$ is

$$\beta \;:=\; \inf_{A} \; \beta_{\mathscr{P}}(A) \;\geq\; \mu_{\min} \;>\; 0,$$

the infimum over all $\mathscr{P}$-realisable regions of their boundary thickness. By Theorem 3.6 the floor is strictly positive.

**Corollary 3.8** (No zero-width cut)**.** *Under Axiom 1, a bounded finite-resolution agent cannot divide a connected whole into a region and its complement by a separator of zero content. Every division it performs deposits content $\geq \beta$ in the separator.*

*Proof.* Immediate from Theorem 3.6 and Theorem 3.7. $\square$

### 3.4 Stability and the role of boundedness

The next two results show the bound is robust and that boundedness is essential, not decorative.

**Proposition 3.9** (Refinement lowers but never abolishes the floor)**.** *Let $\mathscr{P}' \preceq \mathscr{P}$ be a refinement with $\mu'_{\min} \leq \mu_{\min}$. Then $\beta' \leq \beta$ for the refined floor, and still $\beta' \geq \mu'_{\min} > 0$. The floor decreases monotonically under refinement but is bounded below by the finest realisable cell and never reaches zero at any finite stage.*

*Proof.* A finer partition can only shrink the separating cells, so $\beta'_{\mathscr{P}'}(A) \leq \beta_{\mathscr{P}}(A)$ for each $A$, giving $\beta' \leq \beta$. By Theorem 3.6 applied to $\mathscr{P}'$, $\beta' \geq \mu'_{\min} > 0$. By (BRS2) every finite stage carries a positive resolution and hence a positive $\mu'_{\min}$; only in the (non-realisable) limit $\delta \to 0$ would the bound vanish, and that limit is excluded for any embedded agent. □

**Proposition 3.10** (Boundedness is necessary)**.** *The conclusion of Theorem 3.6 fails without (BRS1). On an unbounded space $\mathbb{R}^m$ with Lebesgue measure there exist regions (e.g. half-spaces) whose topological boundary is a hyperplane of $\mathcal{L}$-measure zero, and one may realise separators of arbitrarily small measure by tapering; moreover, with infinite total measure Theorem 2.8 fails and a partition into cells of vanishing diameter at uniform cost is not excluded.*

*Proof.* Take $A = \{x_1 \leq 0\} \subset \mathbb{R}^m$. Then $\partial A = \{x_1 = 0\}$ has $\mathcal{L}(\partial A) = 0$, so the topological separator is sharp. The finite-resolution obstruction of Theorem 3.6 relied on Theorem 2.8, whose proof used $\mu(\Omega) < \infty$; with $\mu(\mathbb{R}^m) = \infty$ the measures of infinitely many cells can sum to $\infty$ without forcing $\inf_i \mu(\mathsf{c}_i) = 0$, so the uniform lower bound $\mu_{\min} > 0$ need not hold. Hence the floor argument does not apply, witnessing the necessity of (BRS1). □

*Remark* 3.11 (What the theorem does and does not say)*.* Theorem 3.6 does not assert that the ambient continuum has thick boundaries; a half-space in $\mathbb{R}^m$ has a measure-zero topological boundary. It asserts that a *bounded finite-resolution agent*, who can name only unions of positive-measure cells, cannot realise that measure-zero cut: its sharpest nameable separator has measure $\geq \mu_{\min}$. The thickness is a property of the agent-in-the-world, jointly with boundedness, not of the world in the abstract. This is the precise sense in which "items are bounded": the divisions by which items are individuated are bounded below in content.

## 4 Individuation by negation, and its non-instantaneity

Before a separator can have thickness there must be something to separate. This section settles how a part comes to be at all in a whole with no distinguished element, and proves two facts that drive the rest of the paper: individuation is forced to proceed by negation (Theorem 4.3), and no individuation that yields a genuine part can be completed in a single step (Theorem 4.6), so that every individuation leaves an untouched residue.

### 4.1 Individuation without a selector

**Definition 4.1** (Individuation)**.** An *individuation* of a part from the whole $\Omega$ is a determination of a region $A$ together with a separator certifying $A$ as distinct from $\Omega \setminus A$. A *selector* is a rule that fixes $A$ by naming an element or distinguished point of $A$ in advance of the separation.

**Definition 4.2** (Negation-sequence)**.** For a region $A \subseteq \Omega$, its *negation-sequence* is its complement together with the separator that certifies the complement,

$$\mathcal{N}(A) \;:=\; \Omega \setminus A,$$

the totality of what $A$ is not. We say $A$ is *individuated by negation* if $A$ is determined as the unique region whose negation-sequence is $\mathcal{N}(A)$; that is, $A = \Omega \setminus \mathcal{N}(A)$, with $\mathcal{N}(A)$ given first.

The point of Theorem 4.2 is that $\mathcal{N}(A)$ requires no selector: it is "everything else," which is completely specified by the whole and the act of withholding $A$, with no element singled out. We now show this is the only selector-free route.

**Theorem 4.3** (Negation is the unique selector-free individuation)**.** *Let $\Omega$ contain no distinguished element (no point, region, or order is given in advance). Then any individuation of a part $A$ that uses no selector determines $A$ as $\Omega \setminus \mathcal{N}(A)$ for its negation-sequence $\mathcal{N}(A)$; and conversely every $\mathcal{N}(A)$ determines a unique such $A$. No selector-based rule individuates a part without invoking a further selector.*

*Proof. (Necessity.)* Suppose an individuation fixes $A$ without a selector. To fix $A$ is to decide, of each part of $\Omega$, whether it belongs to $A$ or to the remainder. A positive rule—one that fixes $A$ by asserting membership of some element or distinguished feature—names that element or feature; by hypothesis no element or feature is distinguished in advance, so the rule must itself supply one. Supplying a distinguished element is exactly providing a selector, and that selector must in turn be fixed: either it is given in advance (contradicting that none is) or it is fixed by a prior rule, which faces the same dichotomy. The regress terminates only if some rule fixes $A$ without naming any element of $A$. The sole such rule available is the withholding of $A$ from the whole: specifying the totality $\Omega \setminus A = \mathcal{N}(A)$ that $A$ is to be distinct from, and taking $A$ as its complement. This names no element of $A$; it names only "the rest," which the whole already supplies. Hence $A = \Omega \setminus \mathcal{N}(A)$.

*(Sufficiency and uniqueness.)* Given a negation-sequence $\mathcal{N}(A) \subsetneq \Omega$ with $\mu(\Omega \setminus \mathcal{N}(A)) > 0$, the region $A := \Omega \setminus \mathcal{N}(A)$ is determined uniquely as a set, since complementation in $\Omega$ is an involution: $\Omega \setminus (\Omega \setminus \mathcal{N}(A)) = \mathcal{N}(A)$. Thus $\mathcal{N}$ is a bijection between admissible parts and admissible negation-sequences, and each side determines the other without further data. $\square$

*Remark* 4.4 (Why complementation closes the regress). A positive selector is a map "choose an element" that must itself be chosen; in a whole with no distinguished element the choice has no ground and regresses. Complementation is the unique operation that fixes a part using only the whole and the part's own absence—data already present once the whole is given. This is why negation, and not selection, is the primitive of individuation here.

## 4.2 Non-instantaneity

We now show that individuation cannot be punctual. The intuition is exact: a separation that takes no extent leaves the part indistinguishable from the whole, hence is no separation at all. We formalise an individuation as a finite sequence of refinements and prove that a zero-length sequence individuates nothing.

**Definition 4.5** (Separation process)**.** A *separation process* for a target region $A$ is a finite chain of realisable partitions $\mathscr{P}_0 \succeq \mathscr{P}_1 \succeq \cdots \succeq \mathscr{P}_k$ together with, at each stage $j$, a $\mathscr{P}_j$-realisable approximant $A_j$ (a union of cells of $\mathscr{P}_j$), such that $A_0 = \Omega$ (the undifferentiated whole) and the chain is *committing*: each step $j \to j+1$ moves at least one cell out of the approximant's separator, i.e. $\beta_{\mathscr{P}_{j+1}}(A_{j+1}) < \beta_{\mathscr{P}_j}(A_j)$ or $A_{j+1} \neq A_j$. The *length* of the process is $k$, the number of committed steps.

**Theorem 4.6** (Non-instantaneity)**.** *Under Axiom 1 with $\Omega$ connected, no separation process of length $k = 0$ individuates a part: if $k = 0$ then $A_0 = \Omega$ and no region $A$ with $\mu(A) > 0$ and $\mu(\Omega \setminus A) > 0$ has been determined. Consequently every individuation of a genuine part has length $k \geq 1$, and at every stage prior to completion a non-empty separator of measure $\geq \beta$ remains uncommitted.*

*Proof.* At length $k = 0$ the only datum is $A_0 = \Omega$. Then $\Omega \setminus A_0 = \varnothing$ has measure zero, so $A_0$ is not a region in the sense of Theorem 2.2: it fails $\mu(\Omega \setminus A_0) > 0$. No part distinct from the whole has been determined; $A_0$ coincides with $\Omega$. Hence a genuine part requires at least one committing step, $k \geq 1$.

For the residue claim, consider any stage $j < k$ with approximant $A_j$. By Theorem 4.3 the still-pending determination of $A$ is the withholding of $\mathcal{N}(A)$, not yet complete at stage $j$, so $A_j \neq A$ and the realisable separator $\Sigma_{\mathscr{P}_j}(A_j)$ is non-empty. By Theorem 3.6, $\mu(\Sigma_{\mathscr{P}_j}(A_j)) \geq \beta > 0$. Thus at every stage before completion a separator of measure at least the floor remains uncommitted—the untouched residue of the individuation in progress. ☐

*Remark* 4.7 (The residue). Theorem 4.6 is the generative fact behind everything that follows. An individuation is not the flash assignment of a boundary but a process that withholds $\mathcal{N}(A)$ step by step; because it has positive length and the whole against which it withholds is itself only ever finitely resolved, what is left uncommitted at the close is never empty. We call this remainder the *residue* of the individuation. Section 8 shows the residue cannot be removed even in principle—not because it is small, but because removing it would require verifying $A$ against the very totality that constitutes it.

**Corollary 4.8** (Individuation is monotone in commitments)**.** *Along any separation process the number of committed steps is non-decreasing, and a step once committed—a cell once removed from the separator—cannot be un-committed without itself being a further step. Hence the committed-count is a monotone record of the process.*

*Proof.* By Theorem 4.5 each step is committing and the chain of partitions is a refinement chain $\mathscr{P}_j \succeq \mathscr{P}_{j+1}$, which is a one-directional order: re-coarsening to undo a commitment is a distinct act, not an inversion of the order on the original chain. Thus the count of committed steps never decreases along the process; it is taken up again as the central object of Section 9. ☐

# 5 Three derivations of the floor

The Boundary–Thickness Theorem produced the floor $\beta > 0$ from the geometry of separators. We now give two further, independent derivations—one representational, one cardinality-theoretic—and a cost argument showing that driving a separator toward zero content has diverging cost. The three agree: each bounds the same constant from below away from zero, and none can be reduced to the others. This redundancy is the paper's evidence that the floor is structural and not an artefact of one formalism.

## 5.1 The representational derivation

The first derivation asks what it costs to *name* a region, independently of any separator. An embedded agent identifies a region by a finite descriptor drawn from a finite scheme; we show no such scheme distinguishes all regions of a bounded resolvable space exactly, so a positive descriptive residue persists.

**Definition 5.1** (Descriptive scheme)**.** A *descriptive scheme* for a partition $\mathscr{P}$ with $N$ cells is an injection $\rho$ from a set $\mathcal{R}$ of realisable regions into the finite set $\{0,1\}^N$ of cell-indicator strings: $\rho(A)$ is the string whose $i$-th entry is 1 iff $\mathsf{c}_i \subseteq A$. A region $A$ is *exactly named* by $\rho$ if $A = \bigcup\{\mathsf{c}_i : \rho(A)_i = 1\}$.

**Proposition 5.2** (Representational residue)**.** *Under Axiom 1, for any region $A$ that is not $\mathscr{P}$-realisable the descriptive scheme $\rho$ names instead the realisable region $\widehat{A} = \bigcup\{\mathsf{c}_i : \mathsf{c}_i \subseteq A\}$, and the discrepancy $A \triangle \widehat{A}$ lies in the separator $\Sigma_{\mathscr{P}}(A)$, of measure $\geq \beta$. No finite descriptive scheme over $\mathscr{P}$ names every region of $\Omega$ exactly; the unnamed discrepancy has measure at least the floor.*

*Proof.* A scheme over $\mathscr{P}$ has range $\{0,1\}^N$, a finite set, while the regions of $\Omega$ (Borel sets modulo null sets) are uncountable under (BRS3); so $\rho$ cannot be injective on all regions and must collapse some region $A$ onto a realisable $\widehat{A}$. The cells contributing to $A$ only partially lie in $\Sigma_{\mathscr{P}}(A)$: a cell $\mathsf{c}_i$ with $\mathsf{c}_i \subseteq A$ is named, a cell disjoint from $A$ is omitted, and a cell meeting both $A$ and $\Omega \setminus A$ is exactly a separator cell. Hence $A \triangle \widehat{A} \subseteq \Sigma_{\mathscr{P}}(A)$, and where $A$ is non-realisable this symmetric difference is non-null and contained in a separator of measure $\geq \beta$ by Theorem 3.6. ☐

*Remark* 5.3 (A faithful image is not the invariant). Theorem 5.2 says that the best name a finite scheme can give a region is a realisable approximant differing from it by at least a floor's worth of content. A representation that is faithful at the resolution of $\mathscr{P}$ is still not the region itself; the gap is the same $\beta$ that the separator carries. The descriptive and the geometric residues coincide.

## 5.2 The cardinality derivation

The second derivation needs no geometry at all—only counting. It shows that the collection of distinctions an agent might draw outruns any finite stage of representation, so that at each finite stage a positive shortfall remains.

**Proposition 5.4** (Cardinality shortfall)**.** *Under (BRS3), the family of regions of $\Omega$ (Borel modulo null) has cardinality of the continuum, while the regions exactly nameable at any finite stage $\mathscr{P}_n$ number at most $2^{N(\mathscr{P}_n)} < \infty$. Hence at every finite stage the nameable regions are a measure-dense but cardinality-negligible subfamily, and the act of naming any further region requires passing to a strictly finer partition, which by Theorem 3.9 carries its own positive floor $\beta_{n+1} > 0$.*

*Proof.* By Theorem 2.8 each $\mathscr{P}_n$ is finite, so its nameable regions (unions of cells) number $2^{N(\mathscr{P}_n)}$. By (BRS3) the partitions generate the Borel $\sigma$-algebra modulo null sets, whose cardinality is that of the continuum; no finite stage attains it. To name a region not realisable at stage $n$ one refines to a stage $m > n$ at which it becomes a union of cells; by Theorem 3.9 every such stage carries $\beta_m \geq \mu_{\min}(\mathscr{P}_m) > 0$. Thus the floor reappears at each refinement; there is no finite stage of exact, complete naming and no stage with zero floor. ☐

*Remark* 5.5. Theorem 5.4 is a pure counting fact and is logically independent of the measure-geometry of Section 3: even if every separator could somehow be made thin, the finite nameable family could never exhaust the continuum of regions, so a fresh refinement—with its own floor—would always be required. The two derivations corner the floor from disjoint directions.

## 5.3 The cost derivation

The third derivation treats the floor as the limit of an optimisation: it shows that the cost of thinning a separator toward zero content diverges, so a finite agent never reaches the sharp cut. We work with an abstract, monotone cost functional and assume only what any reasonable accounting of refinement must satisfy.

**Definition 5.6** (Refinement cost)**.** A *refinement cost* is a functional $\Gamma$ assigning to each realisable partition $\mathscr{P}$ a value $\Gamma(\mathscr{P}) \in [0,\infty)$ such that

(C1) **Monotone.** $\mathscr{P}' \preceq \mathscr{P} \Rightarrow \Gamma(\mathscr{P}') \geq \Gamma(\mathscr{P})$ (finer costs at least as much);

(C2) **Additive over cells.** splitting a cell of measure $m$ into cells of measures $m_1, \ldots, m_r$ adds $\sum_s g(m_s) - g(m) \geq 0$ to $\Gamma$, for a convex $g\colon (0,\infty) \to \mathbb{R}$ with $g(m) \to +\infty$ as $m \to 0^+$;

(C3) **Resolution-bounded.** achieving boundary thickness $\beta_{\mathscr{P}}(A) = t$ for a fixed region requires a partition whose separating cells have measure $\leq t$, hence each of cost $\geq g(t)$.

**Theorem 5.7** (Cost divergence at the sharp cut)**.** *Let $\Gamma$ be a refinement cost. For a fixed region $A$, the cost of realising boundary thickness $t$ satisfies $\Gamma_t \geq g(t) \to +\infty$ as $t \to 0^+$. Hence no finite-cost separation process attains a sharp ($t = 0$) separator; the infimum of attainable thickness over finite-cost processes is a strictly positive $\beta$.*

*Proof.* By (C3), realising thickness $t$ requires separating cells of measure $\leq t$; by (C2) each such cell costs at least $g(t)$, and since at least one separating cell is present (Theorem 3.3), $\Gamma_t \geq g(t)$. Since $g$ is convex with $g(m) \to +\infty$ as $m \to 0^+$, $\Gamma_t \to +\infty$ as $t \to 0^+$. A finite-cost process has $\Gamma \leq C < \infty$ for some bound $C$, so its attainable thickness is bounded below by $t_* := \sup\{t : g(t) \leq C\} > 0$. Setting $\beta = t_*$ gives a strictly positive floor below which no finite-cost process reaches. □

*Remark* 5.8 (Bounded return forces a halt). The cost derivation has a corollary used in Section 9: because the return from thinning a separator—the additional content correctly assigned—is bounded by the finite measure $\mu(\Omega)$ ((BRS1)), while the cost $g(t)$ diverges as $t \to 0^+$, the marginal return per unit cost falls to zero. A finite agent with any positive cost-per-return threshold therefore halts at a strictly positive thickness. The halt is not a failure of resolve but the rational endpoint of a diverging-cost, bounded-return optimisation.

### 5.4 Agreement of the three

**Theorem 5.9** (The floor is one constant)**.** *For a fixed bounded resolvable space and agent, the geometric floor of Theorem 3.7, the representational residue of Theorem 5.2, and the cost-limited thickness of Theorem 5.7 are all strictly positive and bound the same quantity: each equals or exceeds $\mu_{\min}$ at the agent's finest realisable partition, and each vanishes only in the excluded limit $\delta \to 0$. No one of them can be reduced to zero while the others remain positive.*

*Proof.* All three are controlled by $\mu_{\min}$ at the finest realisable partition. The geometric floor satisfies $\beta \geq \mu_{\min}$ (Theorem 3.6); the representational residue is a separator of measure $\geq \beta \geq \mu_{\min}$ (Theorem 5.2); the cost-limited thickness $t_*$ satisfies $g(t_*) \leq C$ with separating cells of measure $\geq \mu_{\min}$ forced by (BRS2), so $t_* \geq \mu_{\min}$. Each is therefore $\geq \mu_{\min} > 0$, and each $\to 0$ only as $\mu_{\min} \to 0$, i.e. $\delta \to 0$, which (BRS2) excludes for an embedded agent. Hence they rise and fall together and are positive together. □

## 6 Detectability and the sorites

We now turn from the existence of the floor to its consequence for what can be *distinguished*. A region is registered by an agent only if it stands out from its own separator; we make this precise and prove that distinguishability is governed by the ratio of a region's scale to its boundary thickness (Theorem 6.3). The sorites paradox follows as a corollary (Theorem 6.6): there is no sharp count at which an aggregate becomes distinguishable, because distinguishability is emergence across a band of positive width, not the crossing of a point.

### 6.1 Scale and distinguishability

**Definition 6.1** (Scale of a region)**.** The *scale* of a $\mathscr{P}$-realisable region $A$ is its measure $\sigma(A) := \mu(A)$. Its *relative thickness* is the ratio

$$\tau_{\mathscr{P}}(A) \;:=\; \frac{\beta_{\mathscr{P}}(A)}{\sigma(A)} \;=\; \frac{\mu(\Sigma_{\mathscr{P}}(A))}{\mu(A)}.$$

**Definition 6.2** (Distinguishable region)**.** A region $A$ is *distinguishable* (by an agent realising $\mathscr{P}$) if its interior content exceeds its separator content,

$$\mu(A \setminus \Sigma_{\mathscr{P}}(A)) \;>\; 0,$$

equivalently if $A$ is not engulfed by its own separator. We say $A$ is *engulfed* if $A \subseteq \Sigma_{\mathscr{P}}(A)$, i.e. every cell of $A$ also borders the complement.

**Theorem 6.3** (Detectability)**.** *Let $(\Omega, d, \mu)$ satisfy Axiom 1 and be connected, and let $\mathscr{P}$ have floor $\beta$. A $\mathscr{P}$-realisable region $A$ is distinguishable if and only if its scale exceeds its boundary thickness:*

$$A \text{ distinguishable} \iff \sigma(A) > \beta_{\mathscr{P}}(A), \qquad \text{equivalently} \qquad \tau_{\mathscr{P}}(A) < 1.$$

*In particular every region with $\sigma(A) \leq \beta$ is engulfed and not distinguishable: below the floor, a region cannot be told from its own boundary.*

*Proof.* The region decomposes as a disjoint union of its interior cells and its separator cells, $A = (A \setminus \Sigma_{\mathscr{P}}(A)) \sqcup (A \cap \Sigma_{\mathscr{P}}(A))$, so $\sigma(A) = \mu(A \setminus \Sigma_{\mathscr{P}}(A)) + \mu(A \cap \Sigma_{\mathscr{P}}(A))$.

($\Leftarrow$) If $\sigma(A) > \beta_{\mathscr{P}}(A) \geq \mu(A \cap \Sigma_{\mathscr{P}}(A))$, then $\mu(A \setminus \Sigma_{\mathscr{P}}(A)) = \sigma(A) - \mu(A \cap \Sigma_{\mathscr{P}}(A)) \geq \sigma(A) - \beta_{\mathscr{P}}(A) > 0$, so $A$ is distinguishable.

($\Rightarrow$) If $A$ is distinguishable, $\mu(A \setminus \Sigma_{\mathscr{P}}(A)) > 0$, so $\sigma(A) = \mu(A \setminus \Sigma_{\mathscr{P}}(A)) + \mu(A \cap \Sigma_{\mathscr{P}}(A)) > \mu(A \cap \Sigma_{\mathscr{P}}(A))$. Since $A$ has at least one separating cell (Theorem 3.3) and is realisable, its separator cells are among $A$'s cells when $A$ is small; in the boundary case the inequality $\sigma(A) > \beta_{\mathscr{P}}(A)$ is exactly the statement that interior content is present. Finally, if $\sigma(A) \leq \beta$ then since $\beta \leq \beta_{\mathscr{P}}(A)$ would force $\sigma(A) \leq \beta_{\mathscr{P}}(A)$, the interior content $\sigma(A) - \mu(A \cap \Sigma_{\mathscr{P}}(A))$ cannot be positive, and $A \subseteq \Sigma_{\mathscr{P}}(A)$: $A$ is engulfed. □

*Remark* 6.4 (Distinguishability is relative to the separator, not absolute)*.* The theorem locates distinguishability not in the absolute size of $A$ but in the comparison $\sigma(A)$ versus $\beta_{\mathscr{P}}(A)$. A region is silent—fails to register—precisely when its separating boundary is thick relative to it. This is the exact inversion the introduction promised: smallness is never itself the cause of silence; being engulfed by one's own boundary is.

## 6.2 The sorites as a corollary

We model an aggregate as a region grown by accretion of sub-floor grains and ask when it becomes distinguishable. The classical sorites asks for the single grain whose addition first makes the aggregate a heap. We show no such grain exists, and locate the reason in the floor.

**Construction 6.5** (Accretive aggregate)**.** Let $A_0 \subset A_1 \subset A_2 \subset \cdots$ be realisable regions with $A_{n+1} = A_n \cup \mathsf{c}_{i_n}$ formed by adjoining a single cell of measure $\mu(\mathsf{c}_{i_n}) \leq \beta$ (a sub-floor grain). Write $\tau_n := \tau_{\mathscr{P}}(A_n)$ for the relative thickness at stage $n$.

**Corollary 6.6** (No sharp threshold; the sorites dissolves)**.** *For the accretive aggregate of Theorem 6.5:*

(i) ***No single grain is decisive.*** *The transition from engulfed ($\tau_n \geq 1$) to distinguishable ($\tau_n < 1$) cannot be effected by any single sub-floor grain alone: each grain changes $\sigma(A_n)$ by at most $\beta$ and $\beta_{\mathscr{P}}(A_n)$ by a bounded amount, so $\tau_n$ moves continuously and no one step carries it across 1 from a value bounded away from 1.*

(ii) ***Distinguishability is emergence across a band.*** *There is a range of stages over which $\tau_n$ descends through a neighbourhood of 1; the aggregate becomes distinguishable somewhere in this band, whose width is set by the floor $\beta$, not at any point.*

(iii) ***The grain is silent because its boundary is thick relative to it.*** *Each grain has $\sigma(\mathsf{c}) \leq \beta \leq \beta_{\mathscr{P}}(\mathsf{c})$, so by Theorem 6.3 every grain is itself engulfed and individually indistinguishable, while the aggregate clears its boundary only once $\sigma(A_n) > \beta_{\mathscr{P}}(A_n)$.*

*Hence the sorites has no decisive grain not because the predicate is vague in a logical sense but because distinguishability is the crossing of a positive-width band $\tau \approx 1$, and a sub-floor grain cannot, by itself, carry an aggregate across a band wider than its own contribution.*

*Proof.* Each adjunction increases $\sigma$ by $\mu(\mathsf{c}_{i_n}) \le \beta$ and changes the separator by at most the measure of the cells whose bordering status flips, a quantity bounded in terms of $\beta$ by Theorem 3.6. Hence $\tau_{n+1} - \tau_n$ is bounded in magnitude by a quantity controlled by $\beta/\sigma(A_n)$, which is $< 1$ once $\sigma(A_n) > \beta$. A bounded per-step change cannot move $\tau$ across the value 1 from a starting value bounded away from 1, giving (i); the descent of $\tau_n$ through a neighbourhood of 1 occupies a range of stages of width controlled by $\beta$, giving (ii); and (iii) is Theorem 6.3 applied to a single sub-floor cell, for which $\sigma(\mathsf{c}) \le \beta \le \beta_{\mathscr{P}}(\mathsf{c})$ forces $\tau_{\mathscr{P}}(\mathsf{c}) \ge 1$, i.e. engulfment. □

*Remark* 6.7 (Relation to theories of vagueness). The standard responses to the sorites—degree-theoretic (Zadeh, 1965), supervaluationist (Fine, 1975; Keefe, 2000), and epistemicist (Williamson, 1994); see Black (1937); Hyde and Raffman (2018)—all locate the difficulty in the semantics of the predicate. The present account is orthogonal and geometric: it places the difficulty in the object, namely in the positive content of the separator that any distinguishing of the aggregate must resolve. There is no hidden sharp threshold (against epistemicism), no many-valued truth (against degree theories), and no need for admissible precisifications (against supervaluation); there is a band of positive width $\beta$ across which a region emerges from its boundary. The dissolution is a theorem about separators, not a logic of partial truth.

# 7 Locating versus visiting

The results so far concern an agent that *individuates* a region: draws a separator, pays the floor, distinguishes the region from its complement. There is a second, entirely different relation a region can stand in—being *arrived at* by a traversal of the space, with no separator drawn and no agent acting. This section formalises the two relations and proves they obey opposite economies. The distinction is purely measure- and order-theoretic; it names no dynamics. Its consequence is a precise account of why a region can be free to be reached yet costly to single out, and a classification of three invalid inferences that silently exchange the two.

## 7.1 Log-measure, probability, and traversal

We add three abstract quantities, each with a standard home in measure theory and information theory (Cover and Thomas, 2006; Shannon, 1948), and one abstract map.

**Definition 7.1** (Log-measure and probability of a region)**.** For a region $W \subseteq \Omega$ with $\mu(W) > 0$, its *probability* is the normalised measure

$$p(W) \;:=\; \frac{\mu(W)}{\mu(\Omega)} \in (0, 1],$$

and its *log-measure* is

$$S(W) \;:=\; \log \mu(W).$$

A region is *rare* if $p(W)$ is small, equivalently if $S(W)$ is small relative to $\log \mu(\Omega)$.

*Remark* 7.2 (One quantity under two names). $S(W)$ and $\log p(W) = S(W) - \log \mu(\Omega)$ differ by an additive constant independent of $W$. "Small log-measure" and "small probability" are therefore not two correlated attributes of a region but a single attribute under two names. No operation lowers $p(W)$ while holding $S(W)$ fixed, or raises one while fixing the other; they move together by definition. This identity is used in Theorem 7.13 and is the abstract counterpart of the familiar log-count relation in information theory.

**Definition 7.3** (Measure-preserving traversal)**.** A *traversal* of $\Omega$ is a measurable map $T\colon \Omega \to \Omega$ with $\mu(T^{-1}E) = \mu(E)$ for every Borel $E$ (it preserves $\mu$). A point $x$ *visits* a region $W$ *at step* $n$ if $T^n(x) \in W$. The *visiting frequency* of $W$ along $x$ is

$$\bar{f}_W(x) \;:=\; \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{1}_W(T^n(x)),$$

when the limit exists. No agent, separator, or choice appears in this definition.

*Remark* 7.4 (Why this is not "dynamics")*.* A traversal is an iterated measure-preserving self-map—an order-theoretic and measure-theoretic object, the discrete shift along the orbit $x, Tx, T^2x, \ldots$. We use it only through two classical facts: that on a finite-measure space such a map returns almost every point to any positive-measure region (recurrence), and that the visiting frequency, when it exists, equals the region's probability (the ergodic average). Both are theorems of measure theory; neither requires a continuum parameter, a rate, or any external scale.

## 7.2 Two relations to a region

**Definition 7.5** (Visiting and locating)**.** A point *visits* a region $W$ under a traversal $T$ when some iterate $T^n(x)$ lies in $W$ (Theorem 7.3): a fact about the orbit, involving no separator and no agent. An embedded agent *locates* $W$ when it individuates $W$—draws a realisable separator marking $W$ off from $\Omega \setminus W$ to its finite resolution (Theorem 4.1, Theorem 3.1): an act carrying the floor residue. Visiting relates a traversal to a point; locating relates an agent to a region.

*Remark* 7.6 (The relations are categorically distinct)*.* At every step a point occupies exactly one maximally fine position, visited because the traversal took it there, with no agent locating it. Conversely an agent may locate a region no orbit visits within any finite horizon. A region's being visited says nothing about its being located, and vice versa. The rest of the section is the consequence of holding the two apart.

## 7.3 The asymmetry

**Theorem 7.7** (Locating–Visiting Asymmetry)**.** *Let $W$ be a realisable region of $(\Omega, d, \mu)$ under Axiom 1 with $\Omega$ connected, and let $T$ be a measure-preserving traversal. Then:*

(i) ***Locating is costly.*** *Any individuation of $W$ by an embedded agent deposits separator content $\geq \beta > 0$ (Theorem 3.6); the cost does not depend on $p(W)$ and is never zero.*

(ii) ***Visiting is free.*** *Almost every point visits $W$ under $T$, with no agent and no separator; by recurrence on the finite-measure space $\mu(\Omega) < \infty$ the visit occurs, and being a fact about the orbit it carries no individuation cost.*

(iii) ***Only frequency scales with rarity.*** *The visiting frequency satisfies $\bar{f}_W(x) = p(W)$ for an ergodic traversal (and is bounded by the recurrence fraction in general); thus a rare $W$ is visited just as freely as a common one, only* less often*. Rarity is a statement about $p(W)$, never about the cost or possibility of a visit.*

*Proof.* (i) is Theorem 3.6: a realisable separator in a connected bounded finitely-resolved space has measure $\geq \mu_{\min} = \beta > 0$, independently of $\mu(W)$. (ii): on a space with $\mu(\Omega) < \infty$ a measure-preserving $T$ returns almost every point of any positive-measure region to that region (the sets $T^{-k}W$ cannot be pairwise disjoint modulo null sets, since their equal measures $\mu(W) > 0$ would otherwise sum past $\mu(\Omega)$); the visit is a property of the orbit and involves no separator, hence no floor. (iii): the visiting frequency is the average of the indicator $\mathbf{1}_W$ along the orbit, which for an ergodic $T$ equals $\int \mathbf{1}_W \,\mathrm{d}(\mu/\mu(\Omega)) = p(W)$; smallness of the frequency is exactly smallness of $p(W)$, and is independent of (i)–(ii). $\square$

*Remark* 7.8 (The asymmetry in one line). A region may be hard to single out and yet effortlessly reached: it is hard to *locate* a distinguished region among many alternatives, but a traversal passes through it regardless of whether anything locates it, and through a rare region only *less frequently*, never at greater cost. The cost lives entirely on the locating side; the rarity lives entirely on the visiting side; nothing carries one to the other.

### 7.4 The cost of locating among alternatives

Locating a single region costs the floor (Theorem 3.6). Locating one region *among many* costs more, and the excess grows with how rare the target is—a quantitative refinement of detectability.

**Theorem 7.9** (Locating cost grows with the log of the alternatives)**.** *Let $W$ be one region among $K$ pairwise-disjoint realisable alternatives of equal measure that together exhaust a fixed portion of $\Omega$. To single out $W$ from the other $K-1$, an embedded agent performs at least $\lceil \log_2 K \rceil$ binary individuations, each depositing residue $\geq \beta$; hence the total locating cost is at least*

$$\beta \lceil \log_2 K \rceil .$$

*Since the target's probability is $p(W) = 1/K$ of the exhausted portion, the cost grows like $\beta \log_2(1/p(W))$: rarer targets are costlier to locate, while (by Theorem 7.7) no rarer to visit.*

*Proof.* Distinguishing $W$ from $K-1$ equally-sized alternatives is a search whose outcome selects one of $K$ possibilities; any decision procedure resolving one of $K$ equiprobable outcomes makes at least $\lceil \log_2 K \rceil$ binary distinctions (a counting bound: $b$ binary tests distinguish at most $2^b$ cases, so $2^b \geq K$). Each binary distinction is an individuation—a separator between the surviving candidates and the rejected ones—carrying residue $\geq \beta$ by Theorem 3.6. Summing over the $\geq \lceil \log_2 K \rceil$ distinctions gives total residue $\geq \beta \lceil \log_2 K \rceil$. With $p(W) = 1/K$ this is $\beta \lceil \log_2(1/p(W)) \rceil$. □

*Remark* 7.10 (Consistency with the cardinality derivation). Theorem 7.9 refines Theorem 5.4: where the cardinality argument showed that exhausting the regions of $\Omega$ requires unbounded refinement, each stage carrying its own floor, this shows that even *within* a fixed finite family of $K$ alternatives the cost of singling out one member scales with $\log K$. The floor appears once per binary distinction, and the number of distinctions is set by the rarity of the target.

### 7.5 Three invalid transfers

The asymmetry blocks a family of inferences that move an attribute of locating onto visiting, or silently alter the measure that constitutes a region. We state the three that arise in practice and show each is invalid; they share a single form (Theorem 7.14).

**Proposition 7.11** (Naming confers no structural priority)**.** *Let $y = T^n(x)$ with $n \geq 1$, so $y$ is reached by the traversal and has predecessors. Designating $y$ "the starting point" is a locating act (a label) and does not endow $y$ with the structural property of an origin—having no predecessor. Any statement whose hypothesis is "has no predecessor" fails to apply to $y$ whatever $y$ is called.*

*Proof.* "Starting point" has two senses: a bookkeeping origin chosen by the agent, and the structural origin, a point with no predecessor under $T$. A reached point $y = T^n(x)$, $n \geq 1$, has the predecessor $T^{n-1}(x)$, so it meets the first sense by stipulation and fails the second by construction. A statement proved under "no predecessor" reads a property $y$ lacks; the label supplies only the inert first sense. The transfer carries nothing. □

**Proposition 7.12** (Specificity does not entail an arranger)**.** *From "$W$ is highly specific" (rare; small $p(W)$) one cannot infer "some agent located or produced $W$." Specificity is a measure attribute of the region; production is an act of an agent. The first does not entail the second.*

*Proof.* Theorem 7.7(ii) provides a standing counterexample: under any traversal, the position $T^n(x)$ at each step is a maximally fine, individually rare location that no agent located—it is visited, not located. Hence a region can be as rare as one likes while having been singled out by no one. Rarity is $p(W)$; arrangement is agency; rarity does not imply agency. □

**Proposition 7.13** (A region cannot be made probable at fixed log-measure)**.** *There is no operation that makes a rare region the most probable destination while it remains rare. "Most probable" is largest measure and "rare" is smallest measure; by Theorem 7.2 these are the same scale, so the demand is that one region be simultaneously least and most. Any construction appearing to achieve it has enlarged the region—changed $\mu(W)$, hence $S(W)$ and $p(W)$ together—and so has replaced the rare region by a different, common one.*

*Proof.* By Theorem 7.2, $p(W)$ and $S(W)$ are the same scale up to an additive constant; maximising $p(W)$ maximises $\mu(W)$. To keep $W$ rare is to keep $\mu(W)$ small; to make it most probable is to make $\mu(W)$ largest. Requiring both is requiring $\mu(W) = \min$ and $\mu(W) = \max$, impossible unless the family is trivial. A purported construction that "raises the probability of the rare region" has increased $\mu(W)$ and thereby exchanged $W$ for a larger region; the original rare region was not made probable but replaced. □

**Proposition 7.14** (The three transfers share one form)**.** *Theorems 7.11 to 7.13 are instances of one invalid move: an attribute established for the agent's relation to a region (locating: naming, singling out, producing) is carried onto the traversal's relation to the region (visiting), or the measure that constitutes the region is silently changed. Under Theorem 7.7 and Theorem 7.2 each move is blocked, because the two relations obey opposite economies and the constituting measure is the probability.*

*Proof.* Theorem 7.11 carries a label (locating) onto a structural predicate of visiting (having no predecessor). Theorem 7.12 carries an agency claim (production) onto a measure attribute (rarity). Theorem 7.13 alters the measure that is the probability. In each case validity would require visiting to inherit locating's cost-structure, or a region to change its measure while keeping its identity; Theorem 7.7 forbids the former and Theorem 7.2 the latter. □

*Remark* 7.15 (What this section adds). The floor (Sections 3 and 5) priced the act of individuation; this section shows that price is paid by *locating* alone and not by *visiting*, that it scales with the log of a target's rarity (Theorem 7.9), and that the appearance of paradox in problems about rare configurations is produced by transferring the locating price onto visiting (Theorem 7.14). A reader will recognise that many concrete problems about rare states—in which a configuration seems impossible to reach because it is costly to arrange—have exactly this shape; the resolution in each case is the asymmetry proved here, that what is costly to single out may be free to be reached and merely seldom so. The self-referential obstruction of the next section shows why even the locating price can never be driven to zero: the residue it leaves cannot be certified away from within.

# 8 Non-self-verifiability and the irreducible residue

The previous sections show the separator has positive content and that this content silences sub-floor regions. We now reach the deepest obstruction: the separator cannot be *verified* from within. To verify a part is to decide it against the totality that constitutes it—its negation-sequence—and that decision is self-referential, with no consistent value. We prove this by a self-contained diagonal argument (Theorem 8.3), deduce that the irreducible residue is not a locatable set but a self-referential impossibility (Theorem 8.6), and conclude that the floor can only ever be bounded below, never exhibited (Theorem 8.7).

## 8.1 Verification as deciding membership in the negation

Recall (Theorem 4.2) that $A$ is constituted as $\Omega \setminus \mathcal{N}(A)$: its identity is fixed by the totality $\mathcal{N}(A) = \Omega \setminus A$ of what it is not. We make verification precise as the decision that constitutes this identity.

**Definition 8.1** (Verifier)**.** A *verifier* for the parts of $\Omega$ is a map $\mathsf{V}$ that, given a part $X$ and a candidate constituting totality $T$, returns $\mathsf{V}(X,T) \in \{0,1\}$, intended to read 1 iff $X$ is exactly the part constituted by withholding $T$, i.e. iff $X = \Omega \setminus T$. To *verify* $A$ is to evaluate $\mathsf{V}(A, \mathcal{N}(A))$: to decide whether $A$ is the part its own negation-sequence constitutes.

*Remark* 8.2 (Why verification is membership in the negation)*.* By Theorem 4.2, $A = \Omega \setminus \mathcal{N}(A)$ exactly when $A$ and $\mathcal{N}(A)$ partition $\Omega$; deciding this is deciding, of the constituting totality $\mathcal{N}(A)$, whether what it omits is precisely $A$—a question about $A$ posed entirely in terms of the collection $\mathcal{N}(A)$ that is defined as "everything but $A$." Verification of $A$ is thus inseparable from a question of self-membership: $A$ figures, through its own definition, inside the totality against which it is checked.

## 8.2 A self-contained diagonal obstruction

We now prove, without invoking any external impossibility theorem, that no verifier decides every part correctly. The argument is a diagonal one, in the shape of the classical results of Cantor (1891); Russell (1903); Tarski (1936) and the general form of Lawvere (1969), but it is self-contained: it uses only the definitions above.

**Theorem 8.3** (Diagonal obstruction to verification)**.** *There is no verifier* $\mathsf{V}$ *such that, for every part* $X$ *of* $\Omega$*,*

$$\mathsf{V}(X, \mathcal{N}(X)) = 1.$$

*More precisely, suppose a verifier is* self-applicable*: it can take as its candidate part the very specification of its own verifying behaviour. Then the part*

$$D \ := \ \{ \, \textit{parts } X : \mathsf{V}(X, \mathcal{N}(X)) = 0 \, \}$$

(*the parts the verifier declares not constituted by their own negation*) *cannot be consistently assigned a verification value:* $\mathsf{V}(D, \mathcal{N}(D)) = 1$ *and* $\mathsf{V}(D, \mathcal{N}(D)) = 0$ *each entail the other.*

*Proof.* Suppose, for contradiction, a self-applicable verifier $\mathsf{V}$ exists. Form $D$ as stated: $D$ collects exactly those parts $X$ for which the verifier returns 0 on the self-referential query $\mathsf{V}(X, \mathcal{N}(X))$. Now ask the verifier about $D$ itself.

If $\mathsf{V}(D, \mathcal{N}(D)) = 1$, then $D$ is declared constituted by its own negation; but by the defining property of $D$, membership-defining condition for $D$ is $\mathsf{V}(X, \mathcal{N}(X)) = 0$, so $D$ satisfying its own membership condition would require $\mathsf{V}(D, \mathcal{N}(D)) = 0$. Reading $D$ through its own definition therefore forces $\mathsf{V}(D, \mathcal{N}(D)) = 0$, contradicting the assumed value 1.

If instead $\mathsf{V}(D, \mathcal{N}(D)) = 0$, then $D$ meets its own membership condition, so $D$ is among the parts collected by $D$, i.e. $D$ is constituted as the part its negation withholds, which is the assertion $\mathsf{V}(D, \mathcal{N}(D)) = 1$, contradicting the assumed value 0.

Both values are self-refuting, so no consistent value exists for $\mathsf{V}(D, \mathcal{N}(D))$. Hence no self-applicable verifier decides every part; in particular none returns 1 on every part, for the construction of $D$ is available whenever the verifier is total. This is the diagonal obstruction. □

*Remark* 8.4 (Self-application is unavoidable for an embedded agent)*.* The hypothesis of self-applicability is not a technical convenience; it is forced by embedding. An agent that verifies parts of $\Omega$ while being itself a part of $\Omega$ can have its own verifying region presented as a candidate part. The diagonal part $D$ is then a genuine region, and Theorem 8.3 applies. An agent that could stand outside $\Omega$ would escape the argument—but no embedded agent can, and the whole subject of the paper is the embedded agent. The obstruction is therefore a structural feature of verification-from-within, not an artefact of an artificial total verifier.

## 8.3 The residue is not a set

The diagonal obstruction lets us characterise the irreducible residue. One might hope that the un-verified remainder of an individuation is a definite set—small, locatable, perhaps measurable—which a finer effort could in principle exhibit and remove. We show this hope is incoherent: a locatable residue would be verifiable, and the diagonal obstruction forbids verifying it.

**Definition 8.5** (Irreducible residue)**.** The *irreducible residue* of the individuation of $A$ is the obstruction to deciding $\mathsf{V}(A, \mathcal{N}(A))$: the content that an individuation cannot certify about itself. It is *exhibitable* if there is a realisable region $E$ with $\mu(E) > 0$ such that verifying $A$ reduces to verifying $A$ on $\Omega \setminus E$ together with a separate certification of $E$.

**Theorem 8.6** (The irreducible residue is not exhibitable)**.** *Under Axiom 1 with an embedded (self-applicable) agent, the irreducible residue of any individuation is not exhibitable: there is no realisable region $E$ of positive measure whose separate certification completes the verification of $A$. The residue is a self-referential impossibility, not a locatable set.*

*Proof.* Suppose the residue were exhibitable by some realisable $E$ with $\mu(E) > 0$. Certifying $E$ separately means deciding $\mathsf{V}(E, \mathcal{N}(E))$—verifying $E$ as the part constituted by its own negation—and, by hypothesis, doing so completes the verification of $A$. But $E$ is itself a part of $\Omega$, and an embedded agent is self-applicable (Theorem 8.4); the diagonal construction of Theorem 8.3 produces a part on which $\mathsf{V}$ has no consistent value, so no verifier certifies every part, $E$ among them. Hence the supposed certification of $E$ cannot be guaranteed to terminate with a consistent value; the reduction fails. Therefore no such $E$ exists: the residue cannot be localised to a positive-measure set and removed by certifying it. It is the standing impossibility of the self-referential decision, not a region awaiting finer treatment. ☐

**Corollary 8.7** (The floor can only be bounded below)**.** *The resolution floor $\beta$ admits a strict positive lower bound ($\beta \geq \mu_{\min} > 0$, Theorem 3.6) but no exact exhibition as $\mu(R)$ for a definite residue set $R$. Every attempt to write the residue as the measure of a located set $R$ supplies an exhibitable $E = R$, contradicting Theorem 8.6. The floor is known only as the positive quantity below which no individuation can certify itself.*

*Proof.* The lower bound is Theorem 3.6. If $\beta = \mu(R)$ for a realisable $R$ of positive measure, then $R$ exhibits the residue, contradicting Theorem 8.6. Hence no such exhibition exists, and $\beta$ is characterised solely as a positive lower bound on un-certifiable content. ☐

*Remark* 8.8 (The recognition gap). Theorem 8.6 has a reading in terms of the relation between an agent and the part it would verify. To certify $A$ completely, the agent must decide $A$ against the totality $\mathcal{N}(A)$ that constitutes $A$; but the agent, being embedded, is itself within that totality, so a complete certification would require it to occupy both sides of the constituting partition at once—to be at once the part verified and a portion of the totality against which it is verified. The residue is the irreducible gap that keeps the two apart. Closing the gap would collapse the distinction on which verification depends; the floor $\beta$ is the measure of how far the gap can never be closed. This is the precise sense in which the boundary of knowledge is constitutive, not contingent: the un-knowable remainder is the very condition under which anything is individuated at all.

# 9 Non-return and the cessation of inquiry

We close the development with two order-theoretic results. The first (Theorem 9.2) shows that the record of committed individuations is monotone under every traversal, so no process returns to a prior individuation state and resemblance never amounts to identity. The second (Theorem 9.4) shows that, because thinning a separator has diverging cost and bounded return, a finite agent halts at the floor; knowledge thereafter attaches to bounded wholes and never to sub-floor parts. Neither result uses any dynamics: both are statements about the order on committed separations.

## 9.1 The committed record

**Definition 9.1** (Committed record)**.** Fix a refinement chain of realisable partitions $\mathscr{P}_0 \succeq \mathscr{P}_1 \succeq \cdots$. The *committed record* after a finite sequence of individuation steps is the count

$$M \;:=\; \#\{\text{separator cells committed across all steps so far}\} \in \mathbb{N},$$

the number of cells moved out of an approximant's separator over the process (Theorem 4.5, Theorem 4.8).

**Theorem 9.2** (Non-return)**.** *The committed record $M$ is monotone non-decreasing under every traversal of the refinement chain. No traversal returns the record to a previously held smaller value; in particular, a step intended to undo a prior commitment is itself a committed step and increases $M$. Consequently two individuation states with the same configuration of cells but different records are distinct, and resemblance of configuration never entails identity of state.*

*Proof.* By Theorem 4.8 each committing step increases the count of committed separator cells by at least one, and the refinement order $\mathscr{P}_j \succeq \mathscr{P}_{j+1}$ is one-directional: to "undo" a commitment is to perform a further act—a re-coarsening or re-drawing—which by Theorem 4.5 is itself a committing step and so increments $M$. Hence $M$ never decreases; any attempted return raises it. If two states $s, s'$ share a configuration of cells but have records $M(s) \neq M(s')$, then they are distinguished by the record itself: there is a partition of states, namely "record $\leq M(s)$ versus record $> M(s)$," separating them. By the principle that states are identical only when no realisable partition distinguishes them, $s \neq s'$. Resemblance of configuration is therefore compatible with difference of record, and identity requires both. $\square$

*Remark* 9.3 (Resemblance versus identity)*.* Theorem 9.2 is the order-theoretic core of irreversibility for individuation: a process may revisit a configuration that *looks like* an earlier one, but the committed record that accompanies it has strictly grown, so the state is new. The monotone record is the invariant that configuration alone cannot capture. Nothing here invokes any flow or external parameter; the monotonicity is a property of the order on commitments.

## 9.2 Cessation

**Theorem 9.4** (Inquiry cessation)**.** *Let $\Gamma$ be a refinement cost (Theorem 5.6) and let the return of an individuation—the additional content it correctly certifies—be bounded by $\mu(\Omega) < \infty$ ((BRS1)). Then for any agent with a positive threshold on cost-per-return there is a finite thickness $\beta_* > 0$ at which further thinning is rejected: the agent halts at a strictly positive floor and certifies nothing below it. Knowledge so obtained attaches to bounded wholes of thickness $\geq \beta_*$ and never to sub-floor parts.*

*Proof.* By Theorem 5.7, realising thickness $t$ costs $\Gamma_t \geq g(t)$ with $g(t) \to +\infty$ as $t \to 0^+$. The return is bounded above by $\mu(\Omega)$, a finite constant. Hence the marginal return per unit cost, $\mu(\Omega)/g(t)$, tends to 0 as $t \to 0^+$. An agent that continues thinning only while cost-per-return stays below a fixed positive threshold $\theta$ stops at the largest $t$ with $g(t) \le \mu(\Omega)/\theta$; call it $\beta_*$. Since $g$ is finite away from 0, $\beta_* > 0$. Below $\beta_*$ the agent does not refine, so it certifies no region of thickness $< \beta_*$; by Theorem 6.3 regions of scale below the floor are in any case engulfed. Thus knowledge attaches to wholes of thickness $\ge \beta_*$ and to no sub-floor part. □

*Remark* 9.5 (Why knowledge attaches to wholes). Theorem 9.4 gives the final shape of the picture. Individuation is by negation (Theorem 4.3); it is non-instantaneous and leaves a residue (Theorem 4.6); the residue cannot be exhibited or certified (Theorem 8.6); the record of commitments only grows (Theorem 9.2); and the cost of pursuing the residue to zero diverges against a bounded return, so a finite agent rationally halts at a positive floor (Theorem 9.4). What such an agent can know are bounded wholes—regions that clear their own boundary thickness—attributed and reasoned about as units. It cannot know its sub-floor parts, not for want of effort but because below the floor a part is indistinguishable from its boundary and its certification is self-referentially obstructed. The boundary of knowledge is the floor, and the floor is positive.

# 10 Scope and conclusion

## 10.1 What has been proved

From the single hypothesis of a bounded resolvable space (Axiom 1) we have established the following, each from stated definitions:

(1) **Individuation by negation** (Theorem 4.3): in a whole with no distinguished element, a part can be fixed only as the complement of its negation-sequence; this is the unique selector-free individuation.

(2) **Non-instantaneity and residue** (Theorem 4.6): no individuation of a genuine part completes in zero steps, and at every stage before completion a separator of measure $\ge \beta$ remains uncommitted.

(3) **Boundary thickness** (Theorem 3.6): every realisable separator has measure $\ge \mu_{\min} > 0$; no sharp cut exists for an embedded agent. The floor $\beta$ is strictly positive.

(4) **Three agreeing derivations** (Theorem 5.9): the geometric, representational, and cost-limited floors are all positive and bound the same constant.

(5) **Detectability and the sorites** (Theorem 6.3, Theorem 6.6): a region is distinguishable iff its scale exceeds its boundary thickness; the sorites dissolves as emergence across a band of width $\beta$, with no decisive grain.

(6) **Locating versus visiting** (Theorem 7.7, Theorem 7.9): individuating a region is costly and the cost grows with the logarithm of the region's rarity, while a measure-preserving traversal visits the region freely and scales with rarity only in frequency; three invalid inferences transferring the former onto the latter are classified and blocked (Theorem 7.14).

(7) **Non-self-verifiability** (Theorem 8.3, Theorem 8.6): verification is a self-referential decision with no consistent value; the irreducible residue is not an exhibitable set, and the floor can only be bounded below, never displayed.

(8) **Non-return and cessation** (Theorem 9.2, Theorem 9.4): the committed record is monotone, so no process returns to a prior individuation state; thinning a separator has diverging cost and bounded return, so a finite agent halts at the floor and knows only bounded wholes.

These results turn on one constant, the floor $\beta > 0$, reached from six independent directions—topological, representational, cardinality-theoretic, economic, self-referential, and order-theoretic—each shown to bound or price the same quantity.

### 10.2 Scope

The theorems are statements about an abstract bounded resolvable space and an embedded agent that individuates parts within it by negation. We have made no claim about any particular system, and the proofs use only the metric–measure and order structure declared in Section 2. The hypotheses (Axiom 1) are deliberately minimal: boundedness with compact closure, finite resolution, and hierarchical refinability. Where each clause is used has been marked, and the necessity of boundedness is shown by a counterexample (Theorem 3.10). Material that could not be proved at this standard has been omitted rather than asserted.

### 10.3 Conclusion

A bounded whole, finitely resolved, cannot be divided into a part and its complement without depositing positive content into the separator, and that content cannot be certified from within or driven to zero. The part is individuated by what it is not; the separation is never instantaneous and never exact; the residue is irreducible; the record of separations only grows; and a finite agent rationally halts at a positive floor, knowing bounded wholes and not their sub-floor parts. The floor $\beta > 0$ is the single invariant through which individuation, boundary thickness, detectability, verification, and non-return all speak. It is the irreducible boundary of a bounded resolvable space.

## References


Garrett Birkhoff. *Lattice Theory.* American Mathematical Society, Providence, RI, 3 edition, 1967.

Max Black. Vagueness: An exercise in logical analysis. *Philosophy of Science*, 4(4):427–455, 1937.

Georg Cantor. Über eine elementare frage der mannigfaltigkeitslehre. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 1:75–78, 1891.

Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory.* Wiley-Interscience, Hoboken, NJ, 2 edition, 2006.

Brian A. Davey and Hilary A. Priestley. *Introduction to Lattices and Order.* Cambridge University Press, Cambridge, 2 edition, 2002.

Ryszard Engelking. *Theory of Dimensions, Finite and Infinite.* Heldermann Verlag, Lemgo, 1995.

Herbert Federer. *Geometric Measure Theory.* Springer, Berlin, 1969.

Kit Fine. Vagueness, truth and logic. *Synthese*, 30(3–4):265–300, 1975.

Kurt Gödel. Über formal unentscheidbare sätze der principia mathematica und verwandter systeme i. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.

Paul R. Halmos. *Measure Theory.* Van Nostrand, New York, 1950.

Witold Hurewicz and Henry Wallman. *Dimension Theory.* Princeton University Press, Princeton, NJ, 1941.

Dominic Hyde and Diana Raffman. Sorites paradox. In Edward N. Zalta, editor, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, 2018.

Rosanna Keefe. *Theories of Vagueness*. Cambridge University Press, Cambridge, 2000.

John L. Kelley. *General Topology*. Van Nostrand, Princeton, NJ, 1955.

F. William Lawvere. Diagonal arguments and cartesian closed categories. *Lecture Notes in Mathematics*, 92:134–145, 1969.

Pertti Mattila. *Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability*. Cambridge University Press, Cambridge, 1995.

James R. Munkres. *Topology*. Prentice Hall, Upper Saddle River, NJ, 2 edition, 2000.

Bertrand Russell. *The Principles of Mathematics*. Cambridge University Press, Cambridge, 1903.

Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

Alfred Tarski. Der wahrheitsbegriff in den formalisierten sprachen. *Studia Philosophica*, 1: 261–405, 1936.

Alfred Tarski. A lattice-theoretical fixpoint theorem and its applications. *Pacific Journal of Mathematics*, 5(2):285–309, 1955.

Timothy Williamson. *Vagueness*. Routledge, London, 1994.

Lotfi A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.